

Track 3: The Microfilm Projector

Mission 3: Project The Blueprint

Advanced Study of Converging Lens Optical Systems

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Abstract

This report presents a complete geometrical-optics analysis of a thin converging lens system designed to project a microfilm blueprint onto a wall placed 1.50 m from the lens. Using optical power in diopters, the Gaussian thin-lens equation, and transverse magnification, we determine the focal length, the exact microfilm position for sharp focus, the magnification, and the image height. A detailed sign-convention analysis is included to prevent common errors. We complement the mathematics with principal-ray tracing diagrams, real-world projector analogies, and a proposed experimental validation with uncertainty estimates.

Abstract (friendly version)

Mission: Project a tiny microfilm (2.0 cm) onto a wall 1.50 m away using a converging lens with power $P = +10$ D.

We compute:

- f from P .
- Where to place the microfilm (object distance).
- Magnification M and projected height.
- How to insert the microfilm so the image is readable.

Key idea: Object just beyond $F \Rightarrow$ image very far and very big, but inverted.

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Team Contributions

Member Name	Contribution
Hammadi Nibel	Final editing, LaTeX formatting , integration of all sections,simulation and final compilation.
RAKI Meriem	Step-by-step resolution: sign convention, thin-lens equation, magnification, interpretation, projector analogy, and simulation concept.
MESSAHLI Hiba Manel	Background theory, ray-tracing explanations, applications, experimental design, error analysis, and references.
Bendahmane Reham	Numerical calculations and direct equation solving.
Amani Takelait	Numerical calculations and direct equation solving.
BOUHEROUA Imane	Numerical calculations and direct equation solving.

Table 1: Distribution of tasks among team members (edit names as needed).

Note: Colored “Friendly Explanation” boxes are simplified summaries to make the technical content easier to understand.

1 Introduction and Background

1.1 Historical context of microfilm technology

Microfilm technology revolutionized archival storage in the 20th century by enabling libraries and institutions to store large volumes of documents in compact form. Microfilm readers rely on the same optical principles as slide projectors and cinema projectors: a transparency is illuminated from behind, and a converging lens projects a magnified image onto a screen.

1.2 Mission scenario and problem statement

Mission scenario: You discover a microfilm containing a critical building blueprint. The microfilm is too small to read directly; you must project a sharp enlarged image onto a wall.

Given / Find

Given:

- Optical power: $P = +10\text{ D}$.
- Screen distance: $OA' = 1.50\text{ m}$.
- Microfilm height: $AB = 2.0\text{ cm}$.

Find:

- Focal length f .
- Exact object position OA .
- Magnification M .
- Image height $A'B'$.
- Correct microfilm orientation.

1.3 Objectives

The primary aim of this track within the *Project Absolute Light* workshop is to consolidate students' understanding of **thin converging lenses** through an engaging, problem-based scenario. By the end of this track, each student should be able to:

1. Convert between **optical power** (measured in diopters) and **focal length** (measured in metres or centimetres), and understand why diopters are a more practical unit for optometrists and instrument designers.
2. Apply the **thin-lens equation** (also called the Gaussian lens formula or the conjugate relation) with strict adherence to the *algebraic sign convention* to locate an image given the object position and focal length.
3. Compute the **transverse magnification** and interpret its sign and magnitude to determine whether an image is inverted or upright, and whether it is enlarged or reduced.
4. Derive the **physical size** of the projected image from the object size and magnification.
5. Explain the practical consequence of a negative magnification for a real-world application such as projecting microfilm.

Objectives

1. Convert optical power (diopters) into focal length.
2. Apply the thin-lens equation to determine the object distance.
3. Compute transverse magnification and interpret its sign (upright vs. inverted).
4. Determine projected image height and practical orientation instructions.
5. Use ray tracing to verify the algebraic results.
6. Propose an experimental validation procedure and error analysis.

1.4 Notation (Symbols)

Symbol	Meaning	Units
O	Optical center of the lens	–
F, F'	Object-side / image-side focal points	–
f	Focal length (signed)	m or cm
P	Optical power $P = 1/f$	D = m ⁻¹
OA	Algebraic object distance (microfilm to lens)	m or cm
OA'	Algebraic image distance (lens to wall)	m or cm
AB	Object height (microfilm height)	m or cm
$A'B'$	Image height (projected height)	m or cm
M	Transverse magnification	–

Table 2: Main symbols used in Track 3.

Important: Algebraic sign convention

We use the **Cartesian convention** (light travels left $\rightarrow$ right):

- Distances measured to the right are **positive**.
- Distances measured to the left are **negative**.
- Real object (microfilm on the left): $OA < 0$.
- Real image (screen on the right): $OA' > 0$.
- Converging lens: $f > 0$, $P > 0$.

Rule: choose one convention and keep it everywhere.

2 Physical Phenomenon and Theoretical Background

2.1 Geometrical optics and the ray model

Geometrical optics models light as rays that travel straight in uniform media and bend when refraction occurs. Since our lens and distances are much larger than the wavelength of visible light, the ray model accurately predicts image position, size, and orientation.

Geometrical optics

Imagine light is like millions of tiny **straight arrows** that fly forward.

In the same material (like air), the arrows go straight.

When light meets a new material (air to glass, glass to air), the arrows **change direction** (they “turn”). This turning is called **refraction**.

Why can we use this simple arrow idea? Because the “wiggle size” of light (its wavelength) is **super tiny**, and our lens is **much bigger** (centimeters). So, for our projector lens, treating light like straight arrows gives very good answers: where the image forms, how big it is, and whether it flips upside down.

2.2 Converging vs. diverging lenses

A **lens** is a transparent optical element with two curved (or one curved and one flat) surfaces that refracts light. We distinguish two fundamental types:

- **Converging lens (convex)**: parallel rays meet at a real focus F' on the image side ($f > 0$).
- **Diverging lens (concave)**: parallel rays separate as if coming from a virtual focus F on the object side ($f < 0$).



Mental picture

Converging lens: can project a **real image** on a wall.

Diverging lens: alone produces only **virtual images** (no wall projection).

2.3 Thin-lens approximation (what we assume and why)

We assume the lens thickness is negligible compared to $|OA|$, OA' , f . Then refraction is modeled at a single plane through O , and the Gaussian thin-lens equation applies.

Thin-lens equation (two equivalent forms)**Signed distances (Cartesian):**

$$\frac{1}{OA'} - \frac{1}{OA} = \frac{1}{f}.$$

Positive distances (sometimes used): if $s = |OA|$ and $s' = OA'$, then

$$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f}.$$

In this report we compute using the signed form to keep the physics of signs clear.

Fast sign check (before any algebra)

For Track 3:

- Microfilm is on the left $\Rightarrow OA < 0$.
- Wall is on the right $\Rightarrow OA' > 0$.
- Lens is converging $\Rightarrow f > 0$.

If you keep these three signs, you will not get lost.

2.4 Optical power and focal length (meaning + units)Optical power P measures the strength of a lens:

$$P = \frac{1}{f},$$

with f in meters and P in diopters ($D = m^{-1}$). A large P means a short f , which means strong focusing.**Physical meaning of f** If rays enter **parallel** to the optical axis, they meet at F' located at distance f behind the lens. So $f = 10$ cm means the focus is only 10 cm from the lens.**2.5 Three regimes of a converging lens**

- **Object beyond $2F$:** image between F' and $2F'$, real, inverted, reduced.
- **Object between F and $2F$:** image beyond $2F'$, real, inverted, enlarged.
- **Object inside F :** image virtual (same side), upright, enlarged (not projectable).

2.6 Transverse magnification (size + orientation in one number)

Magnification M

Magnification M tells you two things at the same time: **how big** the image is, and **whether it is flipped**.

We define:

$$M = \frac{A'B'}{AB} = \frac{OA'}{OA}.$$

How to read M instantly:

- **Size (magnitude):** $|M| = 14$ means the image is **14 times bigger** than the object.
- **Orientation (sign):** $M < 0$ means the image is **inverted** (up becomes down).

So M answers: “**How big?**” and “**Upright or inverted?**” in one number.

3 Detailed Mathematical Calculations

3.1 Given system parameters

- Optical power: $P = +10$ D.
- Screen distance: $OA' = +1.50$ m = +150 cm.
- Microfilm height: $AB = +2.0$ cm.

3.2 Problem 1: Determine focal length from optical power

Calculation of f

$$P = \frac{1}{f} \quad \Rightarrow \quad f = \frac{1}{P} = \frac{1}{10 \text{ m}^{-1}} = 0.10 \text{ m} = 10.0 \text{ cm}.$$

Result: $f = +10$ cm (converging lens).

3.3 Problem 2: Determine the object distance OA

Use the thin-lens equation:

$$\frac{1}{OA'} - \frac{1}{OA} = \frac{1}{f}.$$

Substitute $OA' = +1.50$ m and $f = +0.10$ m:

$$\frac{1}{1.50} - \frac{1}{OA} = 10.$$

So:

$$-\frac{1}{OA} = 10 - \frac{1}{1.50} = 10 - \frac{2}{3} = \frac{28}{3}.$$

Therefore:

$$OA = -\frac{3}{28} \text{ m} = -0.1071 \text{ m} = -10.71 \text{ cm}.$$

Interpretation (why this is the projector position)

Sign: negative means “left side” (object side).

Distance: 10.71 cm is only 0.71 cm beyond $f = 10$ cm.

Meaning: object near $F \Rightarrow$ image far away $\Rightarrow$ big projection.

3.4 Problem 3: Transverse magnification M

$$M = \frac{OA'}{OA} = \frac{1.50}{-0.1071} \approx -14.0.$$

Meaning of $M \approx -14$

- $|M| \approx 14$: 14 times bigger.
- $M < 0$: inverted.

3.5 Problem 4: Image height and microfilm orientation

$$A'B' = M \cdot AB \approx (-14.0) \cdot (2.0 \text{ cm}) = -28.0 \text{ cm},$$

so $|A'B'| = 28.0 \text{ cm}$.

Orientation instruction

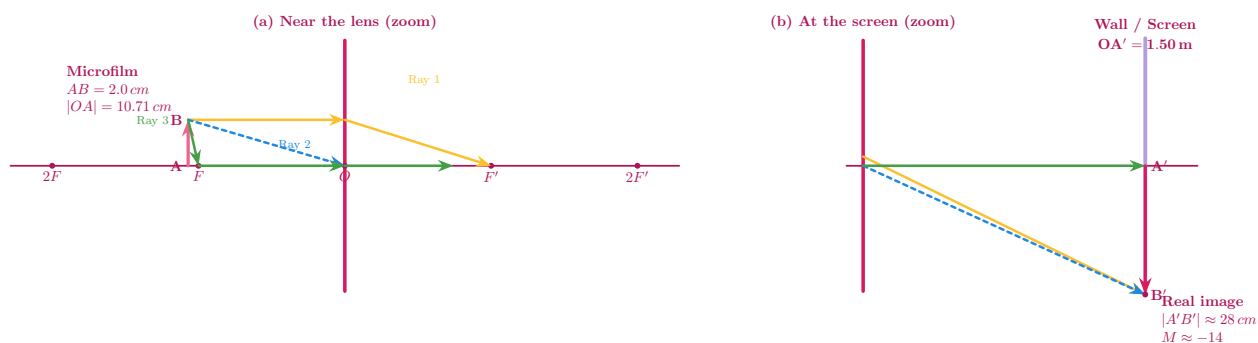
Because $M < 0$, the lens flips the image.

Action: insert the microfilm **upside down** (180° rotation) so the projection reads correctly.

4 Ray Diagram Analysis**4.1 Principal rays (ray tracing rules)**

For a thin converging lens, any two of the following principal rays from the object tip B determine the image tip B' :

1. **Parallel ray:** leaves B parallel to the axis $\rightarrow$ refracts through F' .
2. **Central ray:** passes through O without deviation.
3. **Focal ray:** passes through F $\rightarrow$ emerges parallel to the axis.



Explanation

What the diagram is doing. We take the top point of the object B (microfilm) and trace three **principal rays**. Any two rays are enough to find the image point B' ; drawing all three is a consistency check.

Ray 1 (parallel ray). A ray leaving B parallel to the optical axis hits the converging lens and then refracts through the image-side focal point F' .

Ray 2 (central ray). A ray passing through the optical center O is drawn as a straight line (undeviated in the thin-lens approximation).

Ray 3 (focal ray). A ray aimed toward the object-side focal point F emerges from the lens parallel to the axis.

How this gives a real image. On the right side of a converging lens, the refracted rays physically intersect at one point. That intersection is the image tip B' , so the image is **real** and can be formed on a screen/wall.

Why the image is inverted ($M < 0$). In the diagram, B is above the optical axis but B' is below it. This “top becomes bottom” flip is exactly the sign of a negative magnification.

Why the magnification is large. Here the object is placed just beyond the focal point ($|OA| \approx 10.71$ cm while $f = 10$ cm). This projector configuration produces a large image on a distant wall ($OA' = 1.50$ m).

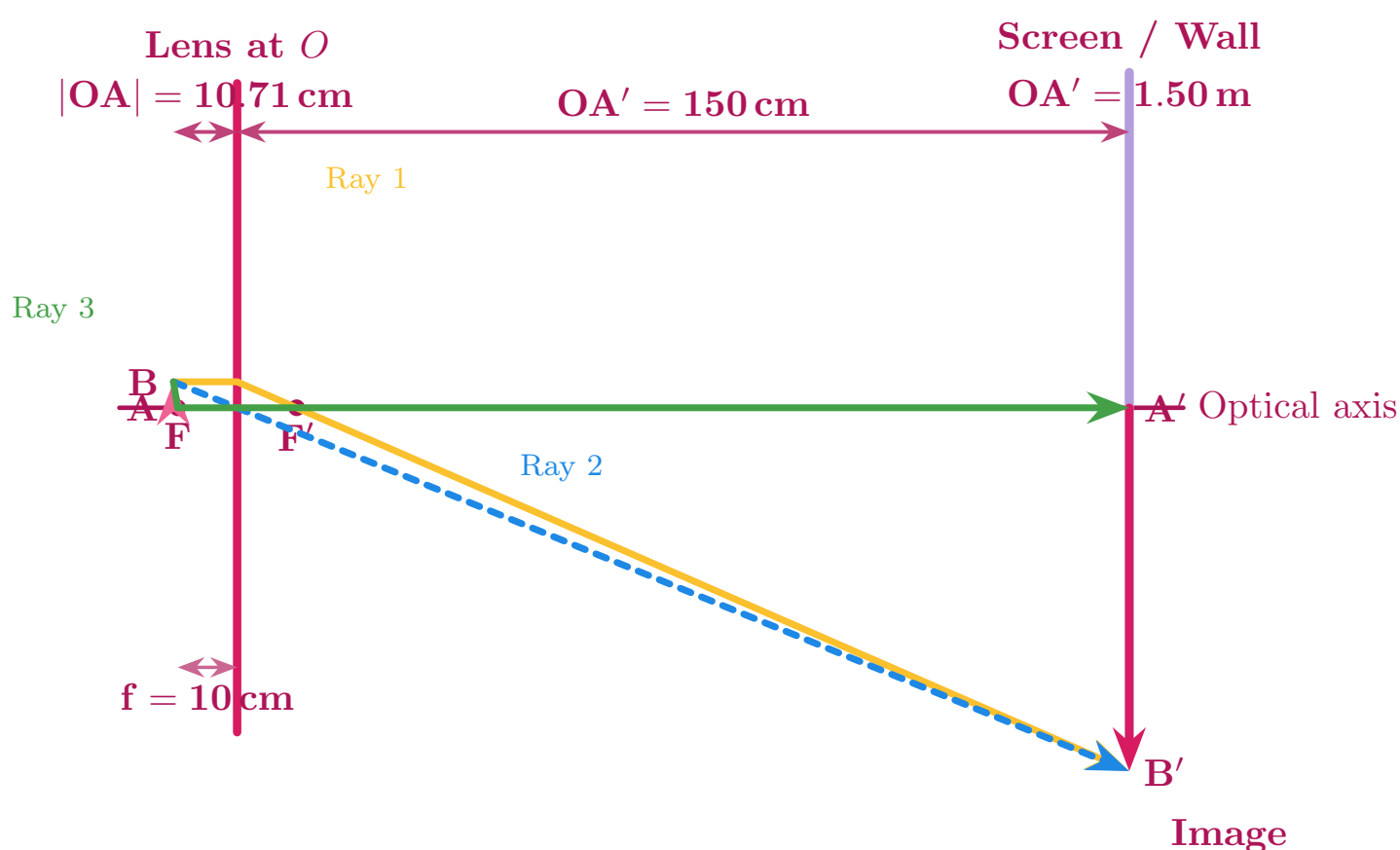


Figure 1: Track 3 Ray Diagram: Full Schematic + Zoomed Construction

5 Real-World Applications and Practical Considerations

5.1 Microfilm readers

Microfilm readers used bright backlighting and a converging lens to project a reduced document onto a readable screen. The same geometry appears in slide projectors: the slide is placed close to the focal point to create a large image on a distant screen.

5.2 Slide and cinema projectors

Professional cinema projectors use multi-element lenses to reduce aberrations, but the basic thin-lens geometry remains: a small transparency is projected onto a large screen, and the film/slide is inserted inverted so the final image is upright.

5.3 Implementation challenges

5.3.1 Illumination requirement

The projected image area scales like M^2 . For $M \approx 14$, area increases by $\approx 14^2 = 196$. This spreads the same light over a much larger area, so strong illumination is required.

5.3.2 Focus precision

The object distance must be close to 10.71 cm. A 1 mm change can blur the image because the object is near the focal plane (high sensitivity). Real devices use fine focusing mechanisms.

5.3.3 Optical aberrations

Real lenses introduce spherical and chromatic aberrations and off-axis defects. Projection lenses use multiple elements to improve sharpness across the whole image.

6 Conclusion

From $P = +10\text{ D}$, the focal length is $f = 10\text{ cm}$. With the wall at $OA' = 1.50\text{ m}$, the thin-lens equation gives $OA = -10.71\text{ cm}$: the microfilm must be placed slightly beyond the focal point on the object side. The magnification is $M \approx -14$, so the image is 14 times larger and inverted, giving $|A'B'| \approx 28\text{ cm}$. Therefore, the microfilm must be inserted upside down so the projected blueprint reads correctly.

Mission 3 checklist (action steps)

1. Put the lens 1.50 m from the wall.
2. Place the microfilm 10.71 cm in front of the lens.
3. Insert the microfilm **upside down**.
4. Illuminate strongly and fine-tune focus by millimeters.
5. Expected projected height: $\approx 28\text{ cm}$.

STATUS: COMPLETE ✓